

Yvette Dixon
CIS 490
Course Project

Project: ADHD Behaviors in Children

Phase 1: Project Question

For this project, I want to explore patterns related to Attention Deficit Hyperactivity Disorder (ADHD) in children. ADHD is a common developmental condition that affects children's learning, behavior, and attention spans. Understanding behavioral patterns associated with ADHD can help people better understand how the condition affects children.

The main question I want to explore is:

Which behavioral characteristics are most associated with ADHD symptoms in children?

I chose this question because ADHD affects many children and can influence school performance and daily activities. By analyzing behavioral patterns, I hope to understand how different traits relate to ADHD symptoms. This topic is meaningful because ADHD can affect children's educational and social relationships, as well as overall development.

Description of Dataset

The dataset I will use is the **ADHD Dataset (4 Classes)** from Kaggle.

Dataset link:

<https://www.kaggle.com/datasets/a7md19/adhd-dataset-4-classes-u2>

This dataset contains information about behavioral characteristics that may be related to ADHD. The dataset includes clean data with each row representing one individual observation. The dataset contains variables such as age, gender, and behavioral indicators related to attention and activity levels. It also includes a class that indicates different levels or types of ADHD-related symptoms.

The dataset is appropriate for this project because it contains more than 3,000 records that can be selected for analysis. The data is organized in a table format that is easy to understand and analyze. This dataset is related to my topic because it focuses on behavioral traits that are connected to ADHD symptoms in children and contains clear variables that can be used to create graphs and summary statistics. The information in the dataset will allow me to examine patterns in ADHD-related behaviors.

Initial Hypothesis

My hypothesis is that certain behavioral characteristics will be strongly associated with ADHD symptoms. I expect that children with higher activity levels and lower attention measures will be more likely to fall into ADHD categories.

I also expect that there may be differences in ADHD patterns based on age and gender. For example, I expect that ADHD-related behaviors may appear more frequently in younger children than in older children. I also expect that boys may appear more often in ADHD categories than girls.

Overall, I expect to find clear patterns showing that certain behaviors are more closely related to ADHD symptoms than others. By analyzing this dataset, I hope to better understand how behavioral characteristics relate to ADHD in children.

Phase 2: Data Cleaning

Before beginning the analysis, the dataset was cleaned and prepared to ensure that the results would be accurate and reliable. Data cleaning is an important step because it helps identify and fix any issues in the dataset, such as missing values, duplicate records, or inconsistent formatting.

Initial Data Inspection

The dataset was first loaded into Python using Google Colab. The structure of the dataset was examined by displaying the first few rows and checking the shape of the data. The dataset contained 6501 rows and 32 columns, meaning there were 6501 observations and 32 variables.

This step confirmed that the dataset was successfully loaded and provided an overview of the available variables, including age, gender, behavioral indicators, and ADHD classification.

Handling Missing Values

The dataset was checked for missing values using the “`.isnull().sum()`” function in Python. This function counts the number of missing values in each column.

The results showed that all columns had zero missing values. This means that there were no gaps or incomplete data in the dataset. Therefore, no action was needed to fill or remove missing values.

Removing Duplicates

The dataset was then checked for duplicate rows using the “*.duplicated().sum()*” function. Duplicate rows can negatively affect analysis by repeating the same information.

The result showed that there were no duplicate records in the dataset. Since no duplicates were found, no rows needed to be removed.

Standardizing Column Names

Next, the column names were standardized to improve consistency and make the dataset easier to work with. All column names were converted to lowercase, and hyphens were replaced with underscores.

For example:

- Q1-1 was changed to q1_1
- Q2-5 was changed to q2_5

This step ensures that the column names are consistent and easier to reference when writing code.

Standardizing Data Formats

Certain columns in the dataset contained categorical (text-based) data, such as:

- family_history
- educational_level
- diagnosis_class

These columns were converted to string format and all values were changed to lowercase. This helps ensure consistency, especially when comparing or grouping data.

The gender column was examined and found to contain numeric values (1 and 2) instead of text labels. Since the data was already standardized in numeric form, no changes were made to this column.

Assumptions and Decisions

Several decisions were made during the data cleaning process:

- It was assumed that the dataset was already clean since no missing values or duplicates were found.
- Column names were standardized to improve readability and consistency.
- Text-based columns were converted to lowercase to avoid inconsistencies in formatting.

- The gender column was left unchanged because it was already encoded numerically.
- Although the dataset contained 6501 rows, it was determined that a subset of the data could be used later if necessary to meet project size requirements.

Final Cleaned Dataset

After completing all cleaning steps, the dataset was saved as a new file named:

cleaned_adhd_data.csv

This cleaned dataset is now ready for analysis in the next phase of the project.

Phase 3: Exploratory Data Analysis

Descriptive statistics were used to understand the basic patterns in the dataset. This included calculating the average (mean), minimum, maximum, and standard deviation for the numeric variables using the “*df.describe()*” function in Python.

The results showed that the average age in the dataset is about 11.5 years old, which fits the focus on children. However, the maximum age is 55, which is unusual and may be an outlier in the data.

The average number of sleep hours is about 7.5 hours, with values ranging from 3 to 12 hours. This shows that there is a wide range of sleep habits among the individuals in the dataset.

Daily phone usage averages about 3.6 hours, but some individuals use their phones up to 10 hours per day. This suggests that some children may have very high screen time.

The focus score ranges from 1 to 10, with an average of about 4.4. This shows that there are clear differences in attention levels, which is important for studying ADHD.

Grouping and Filtering

To better understand patterns in the data, the dataset was grouped by ADHD diagnosis using the “*groupby()*” function in Python. The average values of numeric variables were calculated for each diagnosis group.

The results showed several important patterns. One of the most noticeable differences was in focus scores. Individuals in lower ADHD groups had higher average focus scores (around 5.46), while individuals in higher ADHD groups had lower focus scores (around 3.6). This suggests that decreased focus is associated with higher levels of ADHD symptoms.

Another key finding was difficulty organizing tasks. Individuals in higher ADHD groups had values close to 1.0, while lower ADHD groups had much lower values. This indicates that difficulty with organization is strongly related to ADHD.

Learning difficulties and anxiety/depression levels were also higher in the ADHD groups. This suggests that children with ADHD symptoms may also experience additional challenges related to learning and emotional well-being.

Behavioral question scores (Q1 and Q2 variables) were significantly higher in ADHD groups compared to non-ADHD groups. This confirms that these variables are strong indicators of ADHD-related behavior.

In contrast, variables such as age, sleep hours, and daily activity levels were relatively similar across all groups. This suggests that ADHD differences are more closely related to behavioral and cognitive factors rather than general lifestyle factors.

Overall, grouping the data revealed clear patterns that support the relationship between behavioral characteristics and ADHD symptoms.

Filtering

Filtering was used to focus on a smaller part of the dataset. In this step, the data was filtered to include only individuals with higher ADHD levels (diagnosis classes 2 and 3).

This helped make the analysis easier by focusing only on children who show stronger ADHD symptoms. By looking at this smaller group, it is easier to see patterns and understand how certain behaviors are related to higher ADHD levels.

The filtered dataset still includes the same types of information, such as sleep, activity, focus, and behavior scores, but only for the selected ADHD groups.

Visualization 1 (Bar Chart): Average Focus Score by ADHD Diagnosis

A bar chart was created to show the average focus score for each ADHD diagnosis group. The chart shows that individuals in lower ADHD groups (0 and 1) have higher average focus scores, while individuals in higher ADHD groups (2 and 3) have lower focus scores.

This suggests that as ADHD symptoms increase, focus levels decrease. This is an important finding because it shows a clear relationship between attention and ADHD, which supports the main research question.

Visualization 2 (Histogram): Distribution of Daily Phone Usage

A histogram was created to show how daily phone usage is distributed across individuals. The chart shows that most children use their phones between 2 and 5 hours per day.

There are fewer individuals who use their phones for longer periods, but some values go as high as 10 hours. This suggests that while most children have moderate screen time, a smaller group has very high usage.

This variation in phone usage may be important when examining behavioral patterns related to ADHD.

Visualization 3 (Scatter Plot): Phone Usage vs Focus Score

A scatter plot was created to examine the relationship between daily phone usage and focus score. The chart shows that the data points are spread out, with no clear pattern or trend.

This suggests that there is no strong relationship between phone usage and focus levels in this dataset. Individuals with both high and low phone usage can have a wide range of focus scores.

This indicates that phone usage alone may not be a strong predictor of attention levels and that other factors may play a more important role in ADHD-related behavior.

Summary of Findings

The exploratory data analysis revealed several important patterns related to ADHD in children. The results showed that focus scores decrease as ADHD levels increase, indicating a strong relationship between attention and ADHD symptoms.

Additionally, individuals with higher ADHD levels showed greater difficulty organizing tasks, higher rates of learning difficulties, and increased anxiety levels. These findings suggest that ADHD is associated with multiple behavioral and cognitive challenges.

While screen time varied across individuals, the analysis showed no strong relationship between phone usage and focus scores. This suggests that phone usage alone may not significantly impact attention levels.

Overall, the analysis supports the idea that behavioral characteristics, such as focus and organization, are strongly related to ADHD symptoms.

Method

For this phase, correlation analysis was used to examine the relationship between ADHD diagnosis and other variables in the dataset. This method was chosen because it is simple and helps show how strongly variables are related to each other.

Correlation values range from -1 to 1. A value close to 1 means a strong positive relationship, meaning both variables increase together. A value close to -1 means a strong negative relationship, meaning one variable increases while the other decreases. A value close to 0 means there is little or no relationship.

This method is appropriate for this dataset because it contains many numeric variables related to behavior, focus, and daily habits, which can be compared to ADHD diagnosis.

Results and Interpretation

The correlation analysis showed several important relationships between behavioral variables and ADHD diagnosis.

The strongest relationships were found in the Q1 behavioral variables, which all had correlation values around 0.82 to 0.83. This indicates a very strong positive relationship, meaning that higher scores on these behavioral questions are closely associated with higher ADHD levels.

Difficulty organizing tasks also showed a strong positive correlation of about 0.71. This suggests that children with higher ADHD levels are much more likely to have trouble staying organized.

The Q2 behavioral variables showed moderate to strong correlations, with values around 0.58 to 0.60. These results further support the idea that behavioral patterns are closely related to ADHD.

Learning difficulties had a moderate positive correlation of about 0.49, showing that ADHD is associated with increased challenges in learning.

Focus score showed a negative correlation of about -0.40. This means that as ADHD levels increase, focus decreases. This is an important finding because it directly relates to attention difficulties, which are a key part of ADHD.

Other variables such as sleep hours, physical activity, age, and phone usage showed very weak or no correlation with ADHD. This suggests that these factors may not have a strong impact on ADHD symptoms in this dataset.

Conclusion

The results of the correlation analysis support the original research question, which asked which behavioral characteristics are associated with ADHD symptoms in children.

The findings show that behavioral variables, such as the Q1 and Q2 scores, are strongly related to ADHD. In addition, difficulty organizing tasks and learning difficulties are also important factors. The negative relationship between focus score and ADHD further supports the idea that attention is a key issue in ADHD.

Overall, the results suggest that ADHD is strongly linked to behavioral and cognitive factors rather than lifestyle factors such as sleep or physical activity.

Limitations

There are some limitations to this analysis. First, correlation does not show cause and effect. This means that even though variables are related, we cannot say that one causes the other.

Second, some of the data may include unusual values or outliers, such as age values that are higher than expected for a child dataset. This could affect the results.

Third, some variables may be based on survey responses, which can sometimes be inaccurate or biased.

Lastly, while the dataset is fairly large, it may not represent all children, so the results may not apply to every population.